



Analyzing Failures of Data Augmentation For Low-Resource NER

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EN.705.641 Natural Language Processing: Self-Supervised Models

Common Acronyms and Mnemonics



NER: Named-Entity Recognition

LOC: Location

ORG: Organization

PER: Person

MISC: Miscellaneous

EDA: Easy Data Augmentation

MLM: Masked Language Model

Motivating Example



“Can we augment our way out of a data scarcity problem?”

Often depends on augmentation strategy





Study potential failures of data augmentation

- *How do common augmentation techniques corrupt entity spans and labels?*
- *Do they disproportionately affect certain entity types?*

Propose techniques for addressing data corruption and increasing dataset robustness when augmenting.



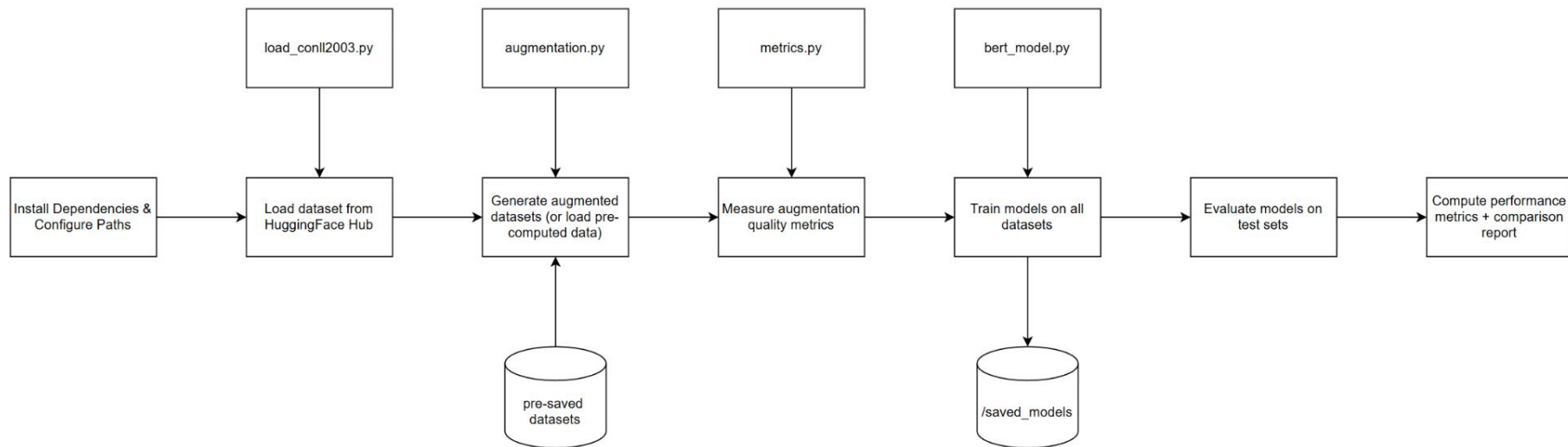
CoNLL-2003 Dataset

Train	14,041 sentences
Validation	3,250 sentences
Test	3,453 sentences
Entity Types	LOC, ORG, PER, MISC
Domain	Reuters news articles
Low-Resource	10% subset

BERT Model

Architecture	bert-base-cased
Task	Token Classification
Fine-Tuning	Full dataset (baseline), 10% subset
Evaluation	Token F1 + Span F1
Framework	HuggingFace Transformers
Metric priority	Span F1 (seqeval)

Experiment Design





“Easy Data Augmentation”: applies synonym replacement, random inserts, swaps, & deletes to **non-entity context words**

*"Goldman Sachs CEO Lloyd Blankfein
met with German Chancellor Angela
Merkel in Berlin on Monday."*



*"Goldman Sachs CEO Lloyd Blankfein
convened with German Chancellor
Angela Merkel in Berlin on Monday."*



Masks context tokens (non-entities) and replace them using BERT's MLM head

"Goldman Sachs CEO Lloyd Blankfein met with German Chancellor Angela Merkel in Berlin on Monday."



*"Goldman Sachs CEO Lloyd Blankfein **held talks** with German Chancellor Angela Merkel in Berlin **last Tuesday**."*



Replaces entity mentions with alternative entities from an external dictionary

"Goldman Sachs CEO Lloyd Blankfein met with German Chancellor Angela Merkel in Berlin on Monday."



"Morgan Stanley CEO Jamie Dimon met with German Chancellor Emmanuel Macron in Paris on Monday."



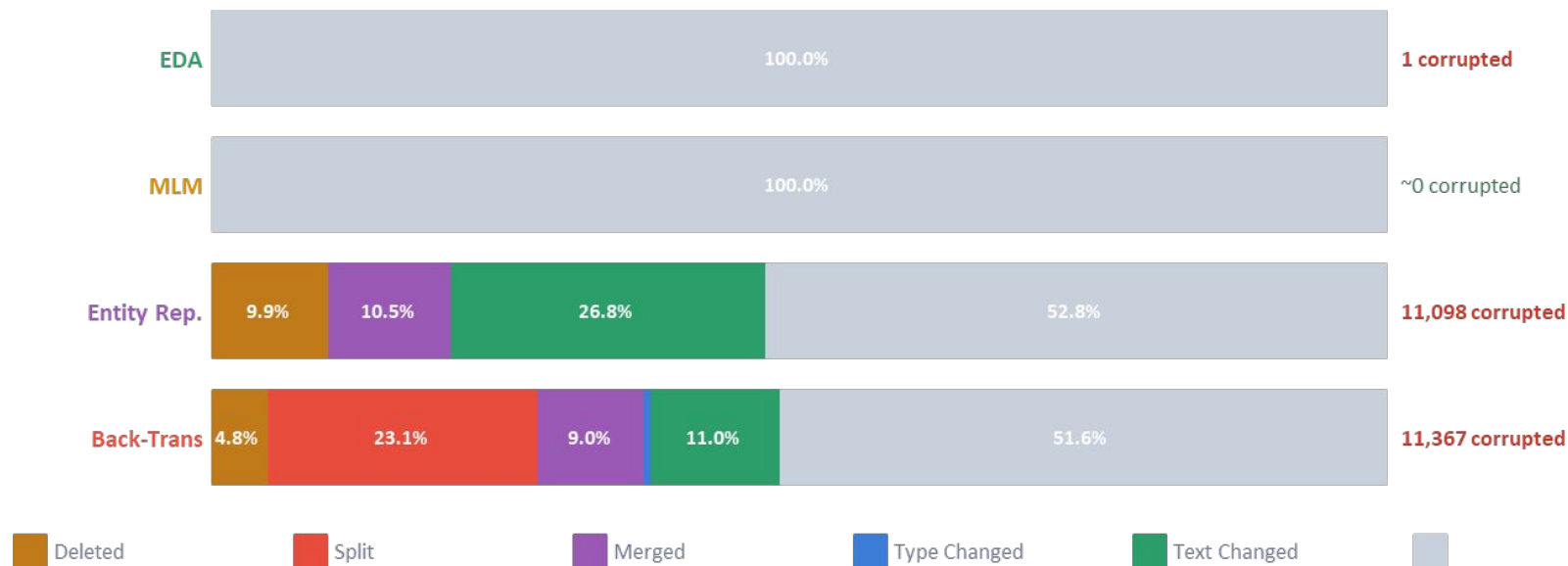
Translates each sentence to an intermediate language and back

"Goldman Sachs CEO Lloyd Blankfein met with German Chancellor Angela Merkel in Berlin on Monday."



"In Berlin on Monday, Lloyd Blankfein of Goldman Sachs met with Chancellor Merkel of Germany."

Span Corruption



Back-Trans: 23.1% of corruptions are SPLITS (fixable). Entity Rep.: 26.8% are TEXT CHANGES (intentional). These failure modes demand different responses.

Token Perturbation



Entity Token Perturbation Rate



Entity replacement perturbs 46.8% of entity tokens yet only drops span F1 by 2pp — BERT's context-based pre-training provides significant tolerance for scoped label noise.

Context (Non-Entity) Token Perturbation

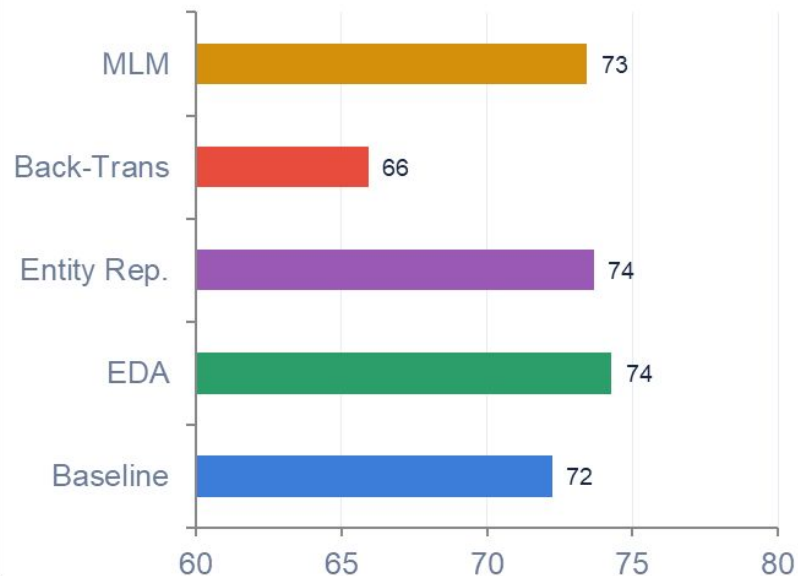


EDA and MLM have nearly identical context perturbation rates (6.6–6.7%) yet EDA outperforms MLM by ~2pp span F1. The gap cannot be explained by noise — it must be in how context is changed.

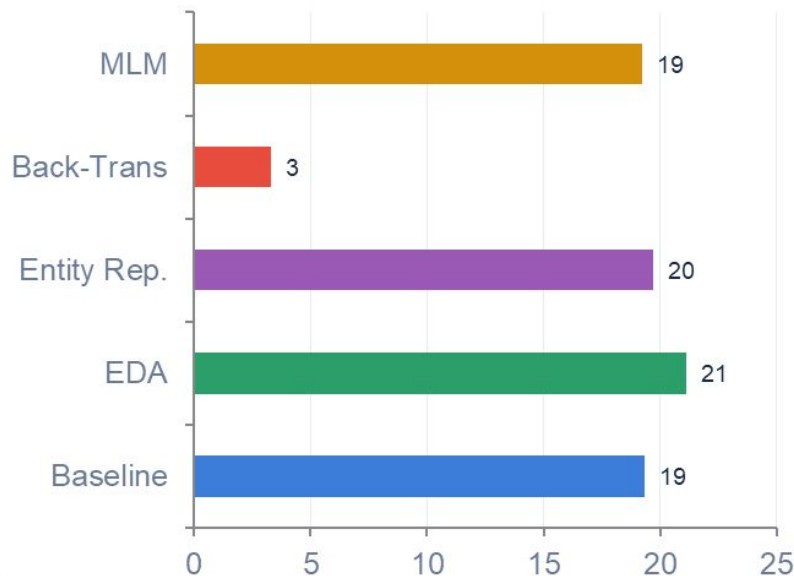
Model Performance



Token-Level F1 — 10% Data (%)



Span-Level F1 — 10% Data (%)



⚠ The token/span divergence is extreme at 10% data — Back-Trans: 65.9% token F1 vs 3.3% span F1, a 62.6pp gap. EDA leads on both metrics.



Potential explanations:

1. Lexical Diversity
2. Distribution Over-Smoothing
3. Semantic Context Shift

Head-to-Head Profile

Metric	EDA	MLM
Span corruption	~0%	0.0%
Label flip rate	~0%	0.0%
Entity token pert.	~0%	0.0%
Context token pert.	6.7%	6.6%
Tokens inserted	4,220	34
Span F1 (10%)	21.1%	19.2%
ORG F1	30.8%	28.8%
PER F1	12.4%	11.1%



1. EDA confirmed virtually noise-free
2. Back-translation is worse than Entity-Aware (though partially fixable)
3. Entity replacement is less destructive than measured
4. Span F1 is the only honest NER metric
5. Two damage architectures revealed
6. EDA vs. MLM paradox: identical profiles, different outcomes



1. Back-Translation + Span Re-Alignment

- a. Take split entity spans and repair them post-augmentation, see if improvements are made.*

2. Ablate EDA vs. MLM Insertion Rate

- a. Run MLM with forced insertions and EDA without insertions.*

3. MISC Learnability Threshold Sweep

- a. Incrementally decrease severity of subsampling to find MISC recovery point, try MISC oversampling to lower the threshold*



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